Problem Set 25

Area of a Surface of Revolution

November 9, 2026

Problem 1

(a)

$$\frac{r_1}{l_1} = \frac{r_2}{l_1 + l}$$
$$r_1 l_1 + r_1 l = r_2 l_1$$
$$\boxed{(r_2 - r_1) l_1 = r_1 l}$$
$$A = \pi r_2 (l_1 + l) - \pi r_1 l_1 = \boxed{\pi (r_1 + r_2) l}$$

(b)

$$l = \sqrt{15^2 + (18 - 10)^2} = 17$$
$$A = \pi (10 + 18) 17 = \boxed{476\pi \text{ cm}^2}$$

(c)

$$8 \cdot l_1 = 10 \cdot 17 \implies l_1 = 85/4$$
$$85/4 \cdot \theta = 2\pi \cdot 10$$
$$R_2 = 17 + 85/4 = 153/4$$
$$\theta = \boxed{\frac{16\pi}{17}}$$

The angle is less than 180°.

Problem 2

(c)

$$y = e^x \implies y' = e^x$$
$$A = \int_0^1 2\pi e^x \sqrt{1 + e^{2x}}\, dx$$
$$= \begin{bmatrix} u = e^x \\ du = e^x\, dx \end{bmatrix} 2\pi \int_1^e \sqrt{1 + u^2}\, du$$
$$= 2\pi \left(\frac{u}{2} \sqrt{1 + u^2} + \frac{1}{2} \ln(u + \sqrt{1 + u^2}) \right) \Big|_1^e$$
$$= \boxed{\pi \left(e\sqrt{1 + e^2} - \sqrt{2} + \ln \left(\frac{e + \sqrt{1 + e^2}}{1 + \sqrt{2}} \right) \right)}$$

Problem 3

(b)

$$y = \sin \pi x \implies y' = \pi \cos \pi x$$

$$A = 2\pi \int_0^1 \sin \pi x \sqrt{1 + \pi^2 \cos^2 \pi x}\, dx$$

$$= \begin{bmatrix} u = \pi \cos \pi x \\ du = -\pi^2 \sin x\, dx \end{bmatrix} \frac{2}{\pi} \int_{-\pi}^{\pi} \sqrt{1 + u^2}\, du$$

$$= \frac{4}{\pi}\left(\frac{u}{2}\sqrt{1+u^2} + \frac{1}{2}\ln(u + \sqrt{1+u^2})\right)\Big|_0^{\pi}$$

$$= \boxed{2\sqrt{1+\pi^2} + \frac{2}{\pi}\ln(\pi + \sqrt{1+\pi^2})}$$

(d)

$$x = y^3 \implies x' = 3y^2$$

$$A = 2\pi \int_1^2 y^3 \sqrt{1 + 9y^4}\, dy$$

$$= \begin{bmatrix} u = y^4 \\ du = 3y^3\, dy \end{bmatrix} \frac{2}{3} \int_1^{16} \sqrt{1 + 9y}\, dy$$

$$= \frac{2}{3} \cdot \frac{1}{9} \cdot \frac{2}{3}(1 + 9y)^{\frac{3}{2}}\Big|_1^{16}$$

$$= \boxed{\frac{5\pi}{27}\left(29\sqrt{145} - 2\sqrt{10}\right)}$$

Problem 4

(d)

$$H = R - R\sin\varphi$$

$$A = \boxed{\frac{1}{2}(1 - \sin\varphi)}$$

$$2 \cdot \sin 30^\circ \cdot \frac{1}{2} = \boxed{\frac{1}{2}}$$

(e)

$$\cos\alpha = R/R + h \implies \sin\frac{\pi}{2} - \alpha = R/R + h$$

$$A = \frac{1}{2}\left(\frac{R + h - R}{R + h}\right) = \boxed{\frac{h}{2(R+h)}}$$

No satellite can see half of the Earth.

Problem 5

(a)

$$y = \frac{1}{2}x^2 \implies y' = x$$

$$A = 2\pi \int_0^{\frac{1}{2}} x\sqrt{1+x^2}\,dx$$

$$= \begin{bmatrix} u = x^2 \\ du = 2x\,dx \end{bmatrix} \int_0^{\frac{1}{4}} \sqrt{1+u}\,du$$

$$= \frac{2}{3}u^{\frac{3}{2}}\Big|_1^{\frac{5}{4}}$$

$$= \frac{\pi(5\sqrt{5}-8)}{12}\ \text{m}^2$$

(b)

$$ds = \sqrt{1+(y')^2}\,dx > dx$$

Problem 6

(a)

$$A = 2\pi \int_C (r-y)\,ds$$

$$= 2\pi \int_C r\,ds - 2\pi \int_C y\,ds$$

$$= \boxed{4\pi^2 r^2}$$

(b)

$$\overline{y} = \frac{1}{L}\int_C y\,ds$$

$$S = 2\pi \int_C y\,ds = \boxed{2\pi \overline{y} L}$$

(c)

$$S = 2\pi \cdot \frac{r}{2} \cdot L = \boxed{\pi r l}$$

$$S = 2\pi \cdot 3\pi \cdot 4 = \boxed{24\pi^2\ \text{cm}^2}$$

Problem 7

(a)

$$A = \pi \int_1^b \frac{1}{x^2}\,dx$$

$$= \pi(-\frac{1}{x})\Big|_1^b$$
$$= \pi - \frac{\pi}{b} < \pi$$

(b)

$$y = \frac{1}{x} \implies y' = -\frac{1}{x^2}$$
$$S(b) = 2\pi \int_1^b \frac{1}{x}\sqrt{1 + \frac{1}{x^4}}\, dx$$
$$\frac{1}{x}\sqrt{1 + \frac{1}{x^4}} > \frac{1}{x}$$
$$S(b) > 2\pi \int_1^b \frac{1}{x}\, dx = 2\pi \ln b$$
$$\lim_{b\to\infty} S(b) \to \infty$$

(c) The surface area of the horn can extend to infinity, while the volume will ultimately converge as the thickness of the horn approaches zero.

Problem 8

(a)

$$y = \frac{x^2}{4} - \frac{\ln x}{2} \implies y' = \frac{x}{2} - \frac{1}{2x}$$
$$\sqrt{1 + (y')^2} = \sqrt{(\frac{x}{2} + \frac{1}{2x})^2} = \boxed{\frac{x}{2} + \frac{1}{2x}}$$
$$l = \int_1^e \frac{x}{2} + \frac{1}{2x} = \frac{x^2}{4} + \frac{1}{2}\ln x\Big|_1^e = \boxed{\frac{e^2+1}{4}}$$

(b)

$$S = 2\pi \int_1^e x(\frac{x}{2} + \frac{1}{2x})\, dx$$
$$= \pi \int_1^e x^2 + 1\, dx$$
$$= \pi(\frac{x^3}{3} + x)\Big|_1^e$$
$$= \boxed{\frac{\pi(e^3 + 3e - 4)}{3}}$$

(c)

$$S = 2\pi \int_1^e (\frac{x^2}{4} - \frac{\ln x}{2})(\frac{x}{2} + \frac{1}{2x})\, dx$$

$$= \frac{\pi}{2} \int_1^e \frac{x^3}{2} + \frac{x}{2} - x \ln x - \frac{\ln x}{x} \, dx$$
$$= \boxed{\frac{\pi(e^4 - 9)}{16}}$$

(d)

$$V = 2\pi \int_1^e x(\frac{x^2}{4} - \frac{\ln x}{2}) \, dx$$
$$= \pi \int_1^e \frac{x^3}{2} - x \ln x \, dx$$
$$= \boxed{\frac{\pi(e^4 - 2e^2 - 3)}{8}}$$

(e)

$$ds > dx$$

Problem 9

(a)

$$c = \frac{e^x + e^{-x}}{2} \implies c' = \frac{e^x - e^{-x}}{2}$$
$$1 + (c')^2 x = 1 + \frac{e^{2x}}{4} - \frac{1}{2} + \frac{e^{-2x}}{4} = c^2$$

(b)

$$S = 2\pi \int_a^b \left(\frac{e^x + e^{-x}}{2} \right)^2 dx$$
$$V = \pi \int_a^b \left(\frac{e^x + e^{-x}}{2} \right)^2 dx$$
$$\boxed{S = 2V}$$

(c)

$$S = 2v$$
$$2\pi \int_a^b f(x) \sqrt{1 + f'(x)^2} \, dx = 2\pi \int_a^b f(x)^2 \, dx$$
$$\sqrt{1 + f'(x)^2} = f(x)$$
$$\boxed{1 + f'(x)^2 = f(x)^2}$$
$$2f'(x)f''(x) = 2f(x)f'(x)$$
$$f'(x) \neq 0 \implies \boxed{f''(x) = f(x)}$$

(d)

$$S = 2\pi \int_{-1}^{1} \left(\frac{e^x + e^{-x}}{2} \right)^2 dx$$
$$= \frac{\pi}{2} \int_{-1}^{1} e^{2x} + 2 + e^{-2x}\, dx$$
$$= \frac{\pi}{2} \left(\frac{e^{2x}}{2} + 2x - \frac{e^{-2x}}{2} \right) \Big|_{-1}^{1}$$
$$= \boxed{\pi \left(\frac{e^2 - e^{-2}}{2} + 2 \right) \approx 17.68}$$

$$A = 2\pi \cdot 2 \cdot \frac{e^x + e^{-x}}{2}$$
$$= \boxed{2\pi \left(e + \frac{1}{e} \right) \approx 19.39}$$

$$A = \boxed{2\pi \left(\frac{e + e^{-1}}{2} \right)^2 \approx 14.96}$$

$$\boxed{14.96 < 17.68 < 19.39}$$